

SYDNEY TECHNICAL HIGH SCHOOL



HIGHER SCHOOL CERTIFICATE ASSESSMENT TASK 2

MARCH 2013

Mathematics Extension 1

General Instructions

- Working time - 70 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in questions 6 to 13
- Start each question on a new page
- A table of standard integrals is provided at the back of this paper

Total marks - 47

Section 1 - 5 marks

Attempt Questions 1 – 5.
Allow about 7 minutes for this section.

Section 2 - 42 marks

Attempt Questions 6 – 11.
Allow about 63 minutes for this section.

Name : _____

Teacher : _____

Section 1

5 marks

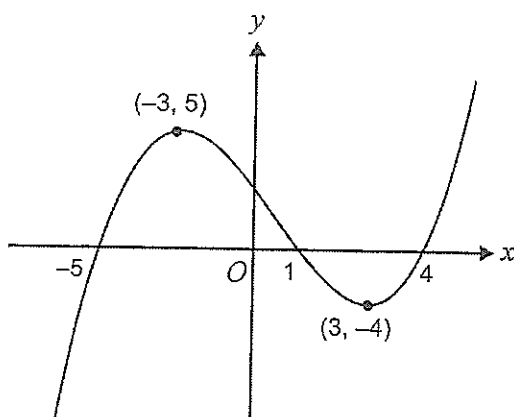
Attempt Questions 1 – 5

Allow about 7 minutes for this section

Use the multiple-choice answer sheet in your answer booklet for Questions 1 – 5.

Do not remove the multiple-choice answer sheet from your answer booklet.

1



For the graph of $y = f(x)$ shown above, $f'(x)$ is negative when

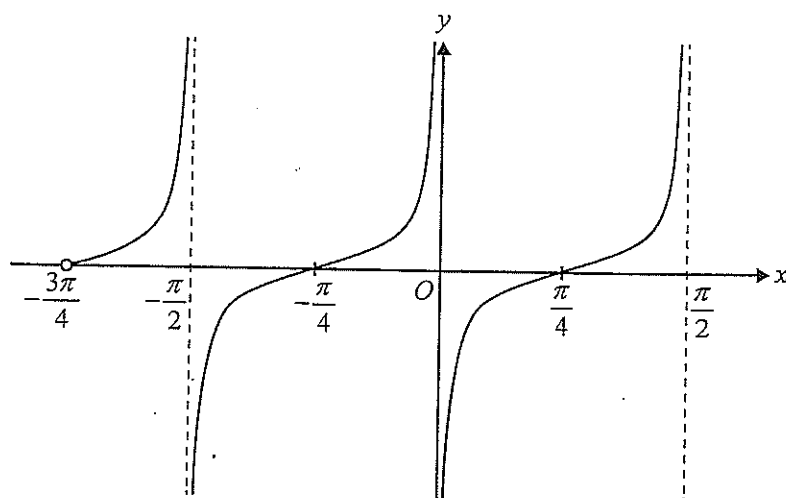
- (A) $-3 < x < 3$
- (B) $x < -3$ or $x > 3$
- (C) $1 < x < 4$
- (D) $-5 < x < 1$ or $x > 4$

2 The volume of the solid of revolution formed by rotating the graph of

$y = \sqrt{9 - (x - 1)^2}$ about the x -axis is given by

- (A) $\pi \int_{-3}^3 (9 - (x - 1)^2) dx$
- (B) $\pi \int_{-2}^4 \sqrt{9 - (x - 1)^2} dx$
- (C) $\pi \int_{-2}^4 (9 - (x - 1)^2) dx$
- (D) $\pi \int_{-2}^4 (9 - (x - 1)^2)^2 dx$

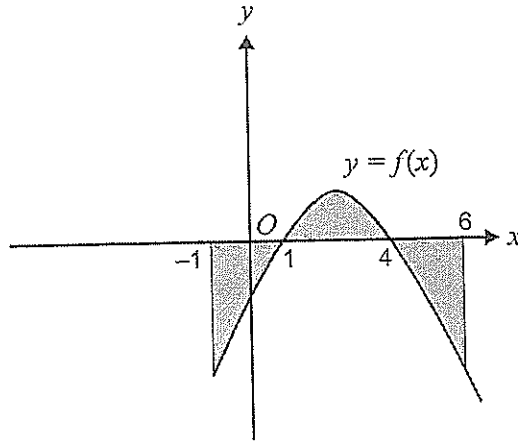
- 3 A section of the graph of f is shown below.



The rule of f could be

- (A) $f(x) = \tan \left(x - \frac{\pi}{4} \right)$
- (B) $f(x) = \tan \left(2 \left(x - \frac{\pi}{4} \right) \right)$
- (C) $f(x) = \tan \left(2 \left(x - \frac{\pi}{2} \right) \right)$
- (D) $f(x) = \tan \left(\frac{1}{2} \left(x - \frac{\pi}{2} \right) \right)$

4



The total area of the shaded regions in the diagram is given by

- (A) $\int_{-1}^6 f(x) dx$
- (B) $\int_1^4 f(x) dx + 2 \int_4^6 f(x) dx$
- (C) $-\int_{-1}^1 f(x) dx + \int_1^4 f(x) dx - \int_4^6 f(x) dx$
- (D) $-\int_{-1}^1 f(x) dx + \int_1^4 f(x) dx - \int_4^6 f(x) dx$

5 If $\frac{d^2y}{dx^2} = x^2 - x$ and $\frac{dy}{dx} = 0$ at $x = 0$, then the graph of y will have

- (A) A maximum turning point at $x = 0$ and a minimum turning point at $x = 1$
- (B) A horizontal point of inflexion at $x = 0$ and $x = 1$, and a minimum turning point at $x = \frac{3}{2}$
- (C) A horizontal point of inflection at $x = 0$, no other points of inflection and a minimum turning point at $x = \frac{3}{2}$
- (D) A horizontal point of inflection at $x = 0$, a minimum turning point at $x = \frac{3}{2}$ and a point of inflexion at $x = 1$

Section 2

42 marks

Attempt Questions 6 – 11

Allow about 63 minutes for this section

Start each question on a new page

Question 6 (7 marks)

- a) Evaluate
- | | |
|---|---|
| i) $\log_9 3$ | 1 |
| | |
| ii) $\lim_{x \rightarrow 0} \frac{4x}{\tan 2x}$ | 1 |
- b) Differentiate $\frac{\sin 2x}{x}$ 2
- c) Find the area bounded by the curves $y = x^2$ and $y = 8x - x^2$. 3

Question 7 (7 marks) Start a new page

- a) Use the substitution $u = 3x - 1$ 3
- to find $\int \frac{x}{(3x-1)^3} dx$
- b) Let $f(x) = \frac{(x+10)^3}{x}$.
- | | |
|---|---|
| i) Find any stationary points of $y = f(x)$ and determine their nature. | 3 |
| | |
| ii) Sketch $y = f(x)$ clearly labelling any important features. | 1 |

Question 8 (7 marks) Start a new page

a) Solve $2 \log(x - 1) - \log(x + 3) = \log 2$ 3

b) i) Show that $\frac{d}{d\theta}(\tan^3 \theta) = 3 \sec^2 \theta (\sec^2 \theta - 1)$ 1

ii) Hence, or otherwise, evaluate $\int_0^{\frac{\pi}{4}} \sec^4 \theta \, d\theta$ 3

Question 9 (7 marks) Start a new page

a) Find a primitive of $\frac{1}{(2x-1)^3}$ 1

b) i) Draw a neat sketch of the curves $y = \cos x$ and $y = \sin 2x$ for $0 \leq x \leq \pi$ on the same diagram. 2

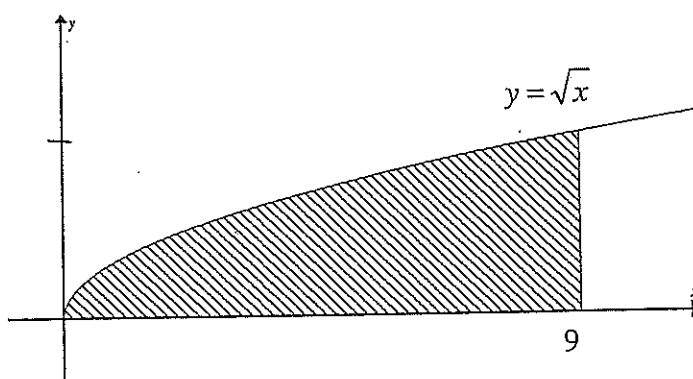
ii) Find the x coordinates of the points of intersection of $y = \cos x$ and $y = \sin 2x$ for $0 \leq x \leq \pi$. 2

iii) Find the area bound by $y = \cos x$, $y = \sin 2x$ and the x -axis for $0 \leq x \leq \pi$. 2

Question 10 (7 marks) Start a new page

a) Find $\int \sin^2 4x \, dx$ 2

- b) Find the volume of the solid formed when the shaded area under the curve $y = \sqrt{x}$, shown below, is rotated about the y axis. 3

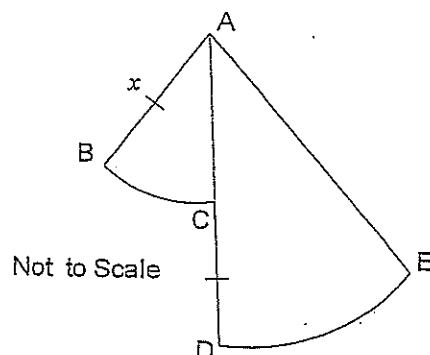


- c) Use Simpson's rule with 3 function values 2

to approximate $\int_0^4 \frac{3}{1 + \sqrt{x}} \, dx$ correct to 2 decimal places.

Question 11 (7 marks) Start a new page

- a) Two sectors make up a company logo as shown below.



Both sectors have centre A, $AB=CD$, $AB=x$ and AC bisects angle BAE.

Let angle $BAC = \theta$ radians.

- i) If the area of the logo is 8π square units, show that $\theta = \frac{16\pi}{5x^2}$ 1
- ii) Find an expression for the perimeter (P) of the logo in terms of x . 2

- b) Evaluate $\int_0^{\frac{\pi}{3}} \frac{\sin 2x}{(1 + \sin^2 x)^2} dx$ using the substitution $u = \sin^2 x$ 4

End of paper

Student Name: _____

Teacher Name: _____

SOLUTIONS - EXT 1 MARCH 2013

1. A

2. C

3. B

4. C

5. D

QUESTION 7

$$a) \int \frac{x}{(3x-1)^3} dx \quad u = 3x-1$$

$$du = 3 dx$$

$$x = \frac{1}{3}(u+1)$$

$$= \frac{1}{9} \int \frac{u+1}{u^3} du$$

$$= \frac{1}{9} \int u^{-2} + u^{-3} du$$

$$= \frac{1}{9} \left[-u^{-1} - \frac{1}{2} u^{-2} \right]$$

$$= -\frac{1}{9} \left[\frac{1}{3x-1} + \frac{1}{2(3x-1)^2} \right] + C$$

QUESTION 6

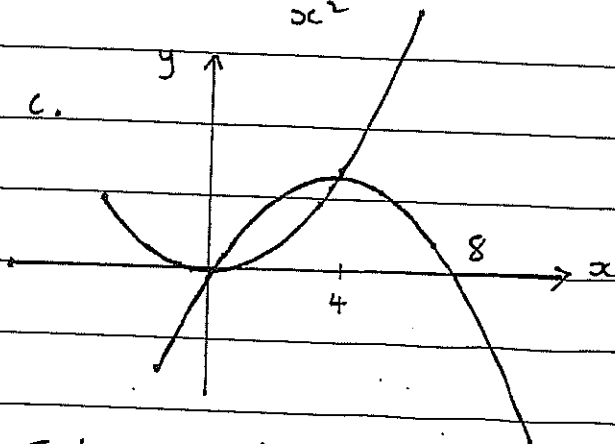
$$a. i) \frac{1}{2}$$

$$ii) 2$$

$$b. \frac{d}{dx} \left(\frac{\sin 2x}{x} \right)$$

$$= \frac{2x \cos 2x - \sin 2x}{x^2}$$

c.



$$b) f(x) = \frac{(x+10)^3}{x^2}$$

$$1) f'(x) = \frac{3x(x+10)^2 - (x+10)^3}{x^2}$$

$$= \frac{(x+10)^2(2x-10)}{x^2}$$

Solve simultaneously.

$$x^2 = 8x - x^2$$

$$2x(x-4) = 0$$

$$x = 0, 4$$

$$\therefore A = \int_0^4 8x - x^2 - x^2 dx$$

$$= \int_0^4 8x - 2x^2 dx$$

$$= \left[4x^2 - \frac{2}{3}x^3 \right]_0^4$$

$$= 21 \frac{1}{3} \text{ sq. units}$$

st. pts when $y' = 0$

$$\text{i.e. } x = -10, 5$$

test $x = 5$

x	4	5	6
y'	-ve	0	+ve

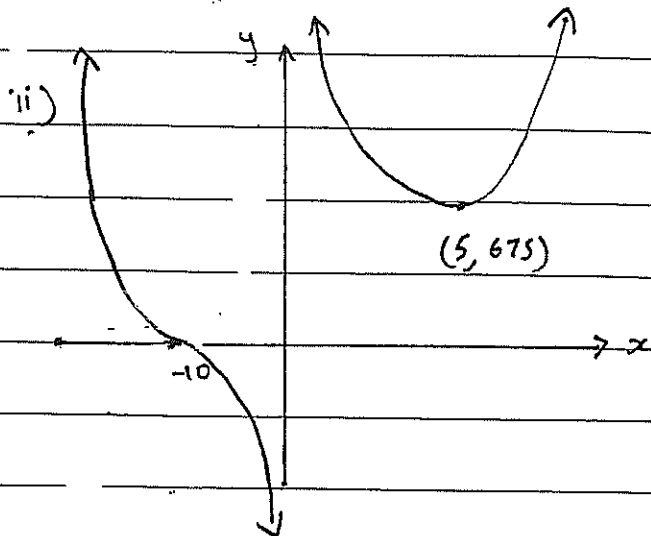
| _ |

 $\therefore$ min at (5, 675)

$$x = -10$$

x	-11	-10	-9
y'	-ve	0	-ve

$\therefore$ horizontal pt. of inflexion at $(-10, 0)$



$$b. i) \frac{d}{d\theta} (\tan^3 \theta)$$

$$= 3 \tan^2 \theta \sec^2 \theta$$

$$= 3(\sec^2 \theta - 1) \sec^2 \theta$$

$$= 3 \sec^2 \theta (\sec^2 \theta - 1)$$

ii)

$$\int_0^{\frac{\pi}{4}} \sec^4 \theta d\theta$$

$$= \frac{1}{3} \left[\int_0^{\frac{\pi}{4}} 3 \sec^2 \theta + \frac{d}{d\theta} (\tan^3 \theta) d\theta \right]$$

$$= \left[\tan \theta + \frac{1}{3} \tan^3 \theta \right]_0^{\frac{\pi}{4}}$$

$$= \left(\tan \frac{\pi}{4} + \frac{1}{3} \tan^3 \frac{\pi}{4} \right) - (0)$$

$$= \frac{4}{3}$$

QUESTION 8

a. $2 \log(x-1) - \log(x+3) = \log 2$

$$\log \frac{(x-1)^2}{x+3} = \log 2$$

$$\frac{(x-1)^2}{x+3} = 2$$

$$x^2 - 2x + 1 = 2x + 6$$

$$x^2 - 4x - 5 = 0$$

$$(x-5)(x+1) = 0$$

$$x = 5, -1 \text{ but } x = -1 \text{ does}$$

not satisfy equation

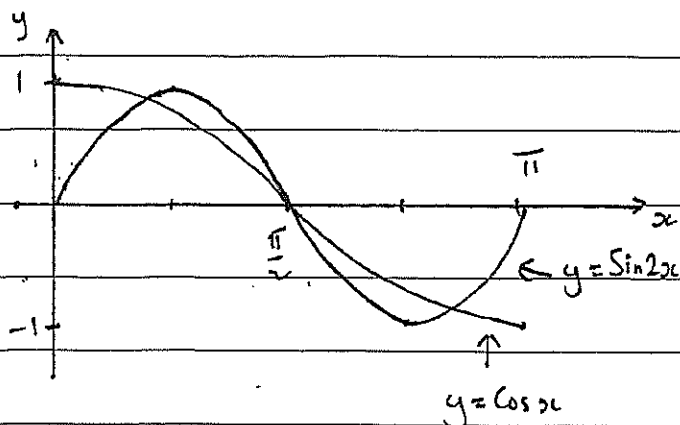
$$\therefore x = 5$$

QUESTION 9

a. $-\frac{(2x-1)^{-2}}{4} + C$

or $-\frac{1}{4(2x-1)^2} + C$

b. i)



$$ii) \cos x = \sin 2x$$

$$\cos x - 2 \sin x \cos x = 0$$

$$\cos x (1 - 2 \sin x) = 0$$

$$\cos x = 0 \quad \sin x = \frac{1}{2}$$

$$x = \frac{\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6}$$

$$iii) A = 2 \int_0^{\frac{\pi}{6}} \sin 2x \, dx +$$

$$2 \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cos x \, dx$$

$$= 2 \left[-\frac{1}{2} \cos 2x \right]_0^{\frac{\pi}{6}} + 2 \left[\sin x \right]_{\frac{\pi}{6}}^{\frac{\pi}{2}}$$

$$= 2 \left(-\frac{1}{2} \cos \frac{\pi}{3} + \frac{1}{2} \cos 0 \right)$$

$$+ 2 \left(\sin \frac{\pi}{2} - \sin \frac{\pi}{6} \right)$$

$$= 2 \left(-\frac{1}{4} + \frac{1}{2} \right) + 2 \left(1 - \frac{1}{2} \right)$$

$$= \frac{3}{2} \approx 1.5 \text{ units}$$

QUESTION 10

$$a) \cos 2A = 1 - 2 \sin^2 A$$

$$\sin^2 A = \frac{1}{2} (1 - \cos 2A)$$

$$\therefore \int \sin^2 4x \, dx$$

$$= \frac{1}{2} \int (1 - \cos 8x) \, dx$$

$$= \frac{1}{2} \left[x - \frac{1}{8} \sin 8x \right] + C$$

$$b) V = \pi \times 9^2 \times 3 - \pi \int_0^3 y^4 \, dy$$

$$= 243\pi - \pi \left[\frac{1}{5} y^5 \right]_0^3$$

$$= 243\pi - \pi \left[\frac{1}{5} \cdot 3^5 - 0 \right]$$

$$= \frac{972\pi}{5} \text{ cubic units}$$

$$c) \begin{array}{c|c|c|c} x & 0 & 2 & 4 \\ \hline y & 3 & 1.243 & 1 \end{array}$$

$$\int_0^4 \frac{3 \, dx}{1 + \sqrt{x}}$$

$$\approx \frac{2}{3} [3 + 1 + 4 \times 1.243]$$

$$= 5.98$$

QUESTION 11

a. i) $A = \frac{1}{2} r^2 \theta$

$$8\pi = \frac{1}{2} x^2 \theta + \frac{1}{2} (2x)^2 \theta$$

$$8\pi = \frac{1}{2} x^2 \theta + 2x^2 \theta$$

$$16\pi = x^2 \theta + 4x^2 \theta$$

$$16\pi = 5x^2 \theta$$

$$\theta = \frac{16\pi}{5x^2}$$

ii) $P = 4x + x\theta + 2x\theta$

$$= 4x + 3x\theta$$

$$= 4x + 3x \left(\frac{16\pi}{5x^2} \right)$$

$$= 4x + \frac{48\pi}{5x}$$

b. $\int_0^{\frac{\pi}{3}} \frac{\sin 2x}{(1 + \sin^2 x)^2} dx$

$$= \int_0^{\frac{3}{4}} \frac{du}{(1+u)^2}$$

$$= \int_0^{\frac{3}{4}} (1+u)^{-2} du$$

$$= \left[-(1+u)^{-1} \right]_0^{\frac{3}{4}}$$

$$= \left(\frac{-1}{1+\frac{3}{4}} \right) - \left(\frac{-1}{1} \right)$$

$$= \frac{3}{7}$$

$$u = \sin^2 x$$

$$du = 2 \sin x \cos x dx$$

$$= \sin 2x dx$$

$$x=0 \quad u=0$$

$$x=\frac{\pi}{3} \quad u=\frac{3}{4}$$